COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CpE212	**SECTION:**	CpE31s2
DATE PERFORMED:	Oct 29, 2026	**DATE SUBMITTED:**	Oct 29, 2026
NAME:	Jeus Guille V. Pizarro	**INSTRUCTOR:**	Engr Jimlord Quejado

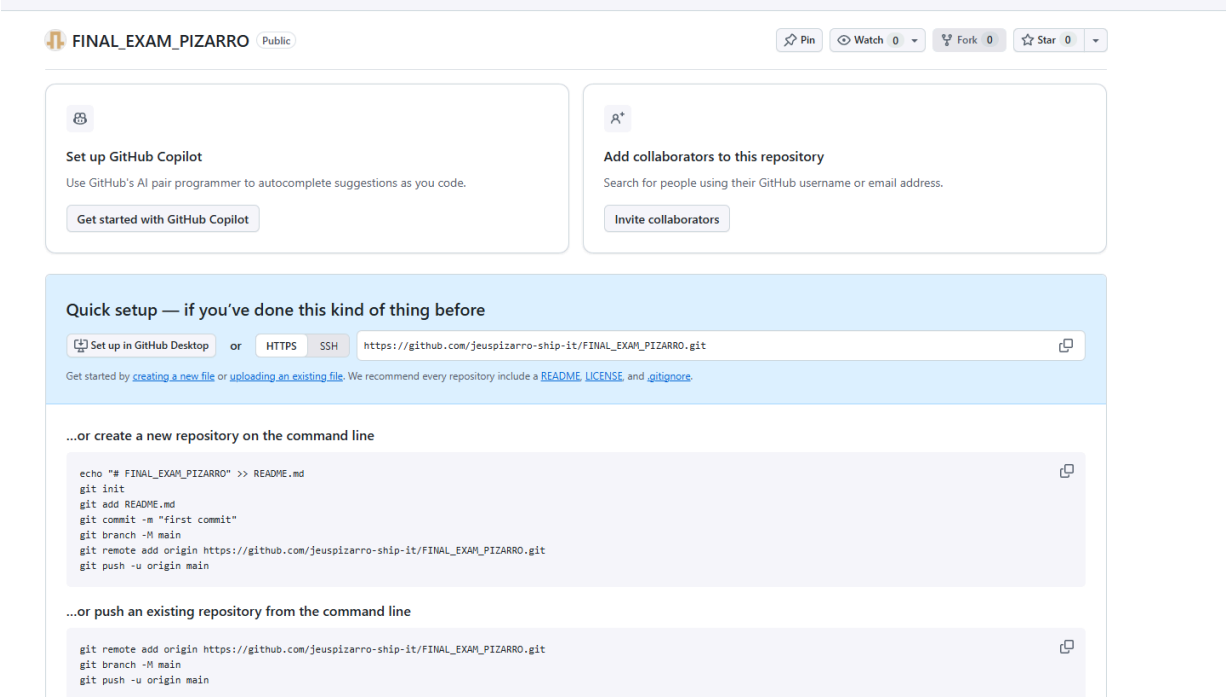
ACTIVITY NO. ___ Final Skills Exam

1. PROCEDURE OUTPUT

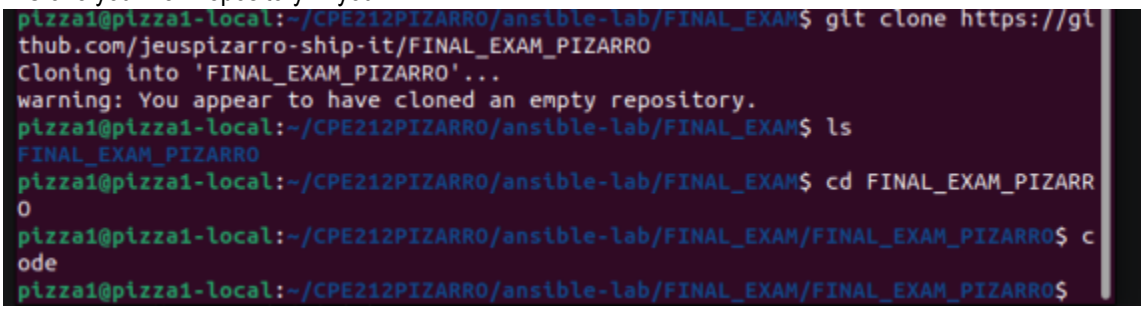
What procedures did you perform?

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

1. Create a repository and label it as "Final_Exam_Surname"

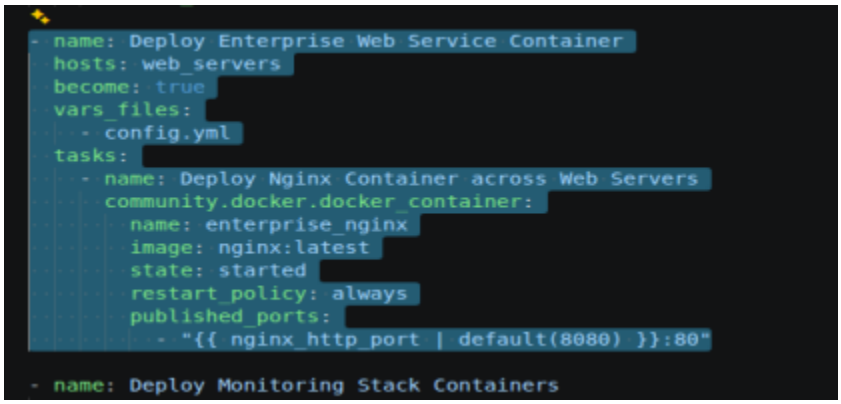


2. Clone your new repository in your VM



3. Create an Ansible playbook that does the following with an input of a config.yaml file and structure inventory file.

3.1 Install and configure one enterprise service that can be installed in Debian and Centos servers

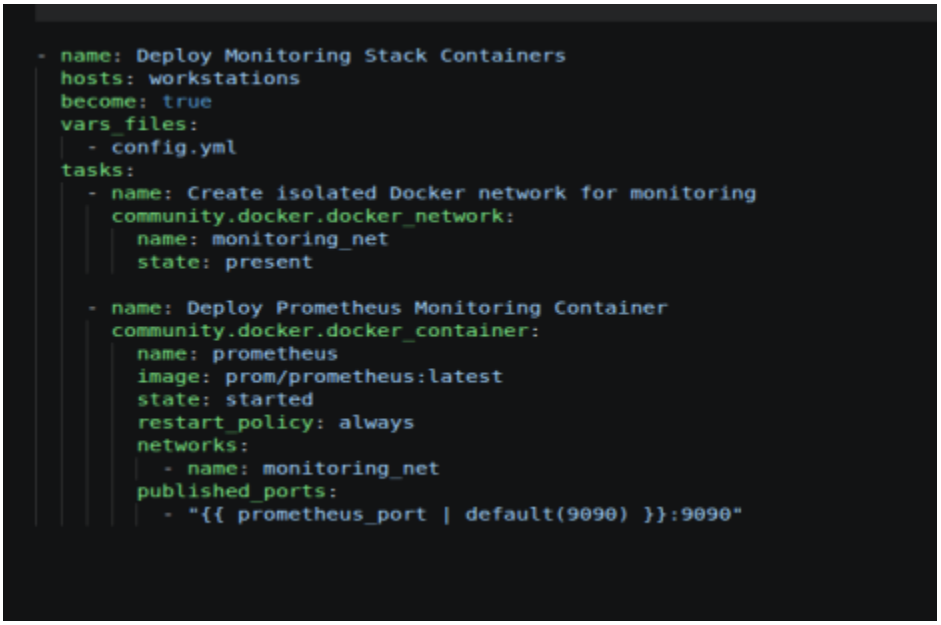


```yaml
- name: Deploy Enterprise Web Service Container
  hosts: web_servers
  become: true
  vars_files:
    - config.yml
  tasks:
    - name: Deploy Nginx Container across Web Servers
      community.docker.docker_container:
        name: enterprise_nginx
        image: nginx:latest
        state: started
        restart_policy: always
        published_ports:
          - "{{ nginx_http_port | default(8080) }}:80"

- name: Deploy Monitoring Stack Containers
```

For this part, i used NGINX for enterprise

3.2 Install and configure one monitoring tool that can be installed in Debian and Centos servers (if it is a stack there should be option of different host)



```yaml
- name: Deploy Monitoring Stack Containers
  hosts: workstations
  become: true
  vars_files:
    - config.yml
  tasks:
    - name: Create isolated Docker network for monitoring
      community.docker.docker_network:
        name: monitoring_net
        state: present

    - name: Deploy Prometheus Monitoring Container
      community.docker.docker_container:
        name: prometheus
        image: prom/prometheus:latest
        state: started
        restart_policy: always
        networks:
          - name: monitoring_net
        published_ports:
          - "{{ prometheus_port | default(9090) }}:9090"
```

For this part, i used Promtheus for monitoring tool

4.4 Change Motd as "Ansible Managed by <username>"

```
6    - config.yml
7  tasks:
8    - name: Change Message of the Day (MOTD)
9      ansible.builtin.copy:
10       content: "{{ motd_text }}\n"
11       dest: /etc/motd
12       owner: root
13       group: root
14       mode: '0644'
15
16   - name: Install Docker prerequisites on Debian hosts
```

Code snippet for motd

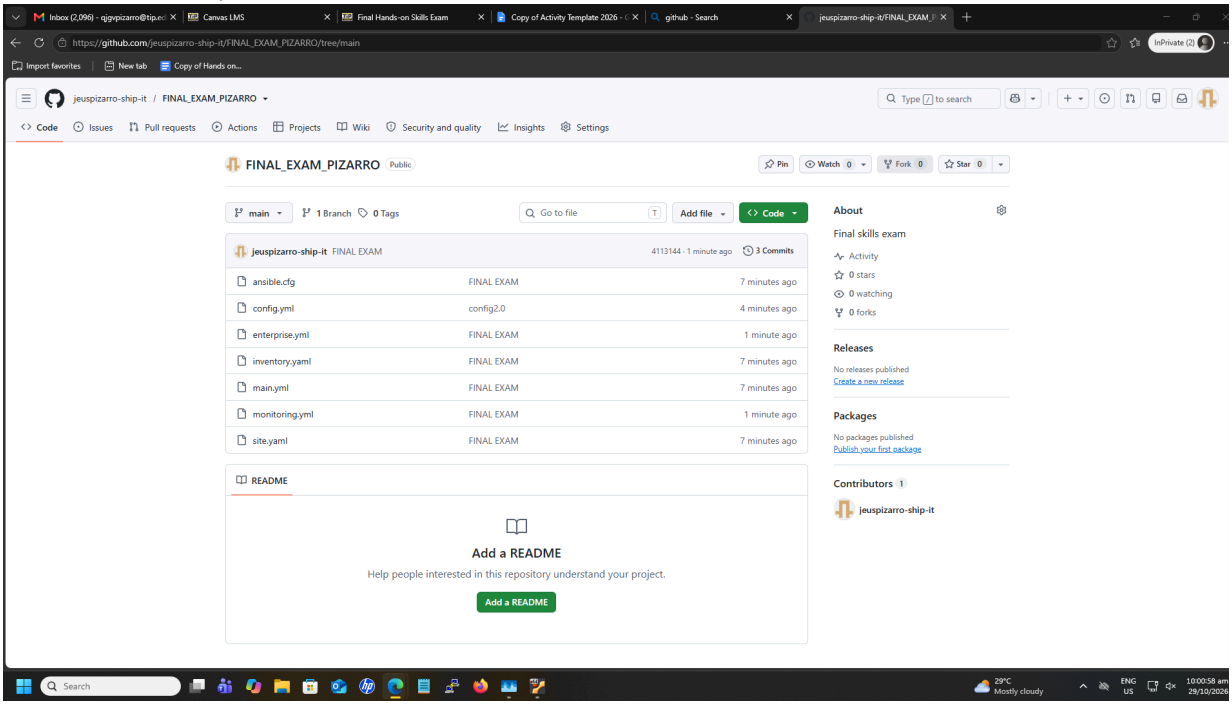
```
pizza2@server1:~/Desktop$ cat /etc/motd
Ansible Managed by PIZARRO
pizza2@server1:~/Desktop$
```

Verification for Motd for ubuntu server

```
[jeus@localhost ~]$ cat /etc/motd
Ansible Managed by PIZARRO
[jeus@localhost ~]$ S
```

Verification for Motd for Centos server

4. Push and commit your files in GitHub



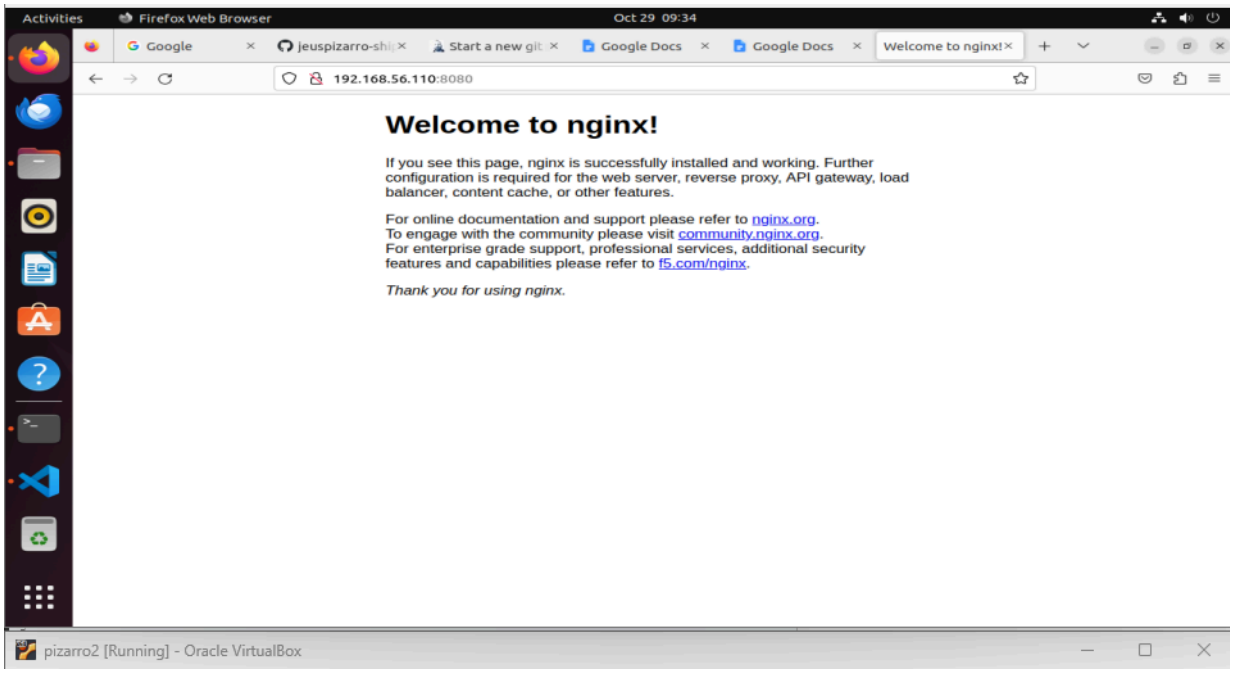
5. Make sure to show evidence of input (codes) process (codes successfully running) and output (evidence of installation)

```
PLAY RECAP *********************************************************************************************************************
pizza2                     : ok=8    changed=0    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
pizza3                     : ok=0    changed=0    unreachable=1    failed=0    skipped=0    rescued=0    ignored=0
pizza_centos               : ok=9    changed=3    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
workstation1               : ok=10   changed=1    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0

pizza1@pizza1-local:~/CPE212PIZARRO/ansible-lab/FINAL_EXAM/FINAL_EXAM_PIZARRO$ s
```

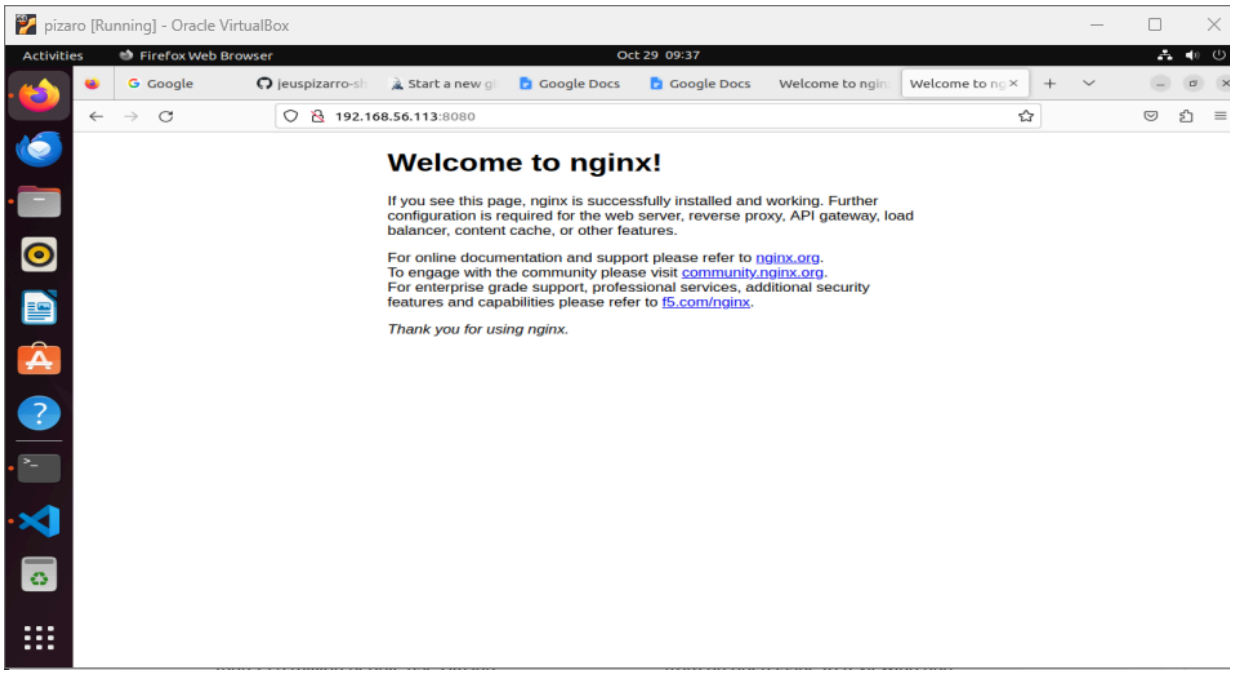
Recap of the code run successfully

```
pizza2@server1:~/Desktop$ sudo docker ps
[sudo] password for pizza2:
Sorry, try again.
[sudo] password for pizza2:
CONTAINER ID   IMAGE                    COMMAND                  CREATED
  STATUS          PORTS                    NAMES
3fb6f41a3ef1   grafana/grafana:latest   "/run.sh"                15 minutes ago
  Up 15 minutes   0.0.0.0:3000->3000/tcp   grafana
c35122d4ac2f   nginx:latest             "/docker-entrypoint.…"   29 minutes ago
  Up 29 minutes   0.0.0.0:8080->80/tcp     enterprise_nginx
cd317507d78f   prom/prometheus:latest   "/bin/prometheus --c…"   35 minutes ago
  Up 35 minutes   0.0.0.0:9090->9090/tcp   prometheus
pizza2@server1:~/Desktop$ s
```

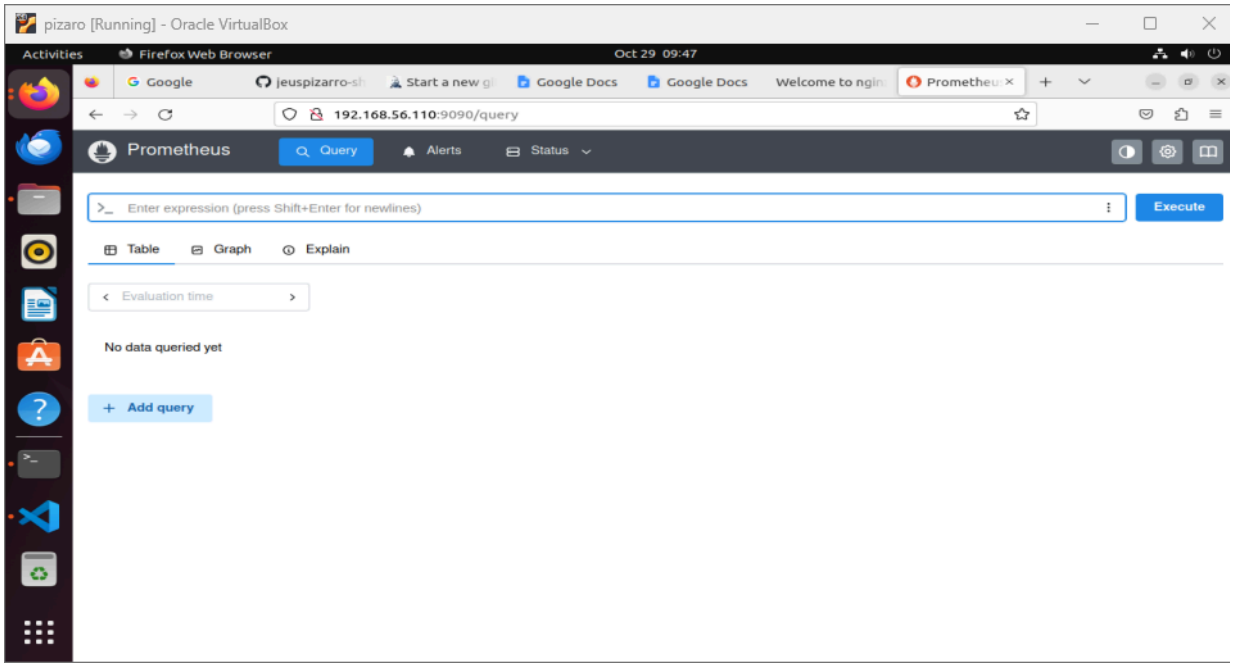


Verification of nginx for ubuntu server

```
[jeus@localhost ~]$ sudo docker ps
[sudo] password for jeus:
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS
       PORTS                    NAMES
d01478cad047   nginx:latest   "/docker-entrypoint.…"   20 minutes ago   Up 20 mi
nutes   0.0.0.0:8080->80/tcp   enterprise_nginx
[jeus@localhost ~]$ S
```



Verification of nginx for centos server



Verification of PROMTHEUS for ubuntu server

5. For your final exam to be counted, please paste your repository link as an answer in this exam.
https://github.com/jeuspizarro-ship-it/FINAL_EXAM_PIZARRO

Note: Extra points if you will implement the said services via containerization.

In this practical skills examination, an automated deployment strategy was successfully executed using an Ansible playbook to configure enterprise and monitoring infrastructure. Key tasks performed included configuring and deploying Nginx web server containers across target nodes, setting up a Prometheus monitoring stack container within an isolated Docker network, and standardizing system messaging by updating the Message of the Day (MOTD) to indicate Ansible management. All code artifacts, playbooks, and configuration files were tracked, committed, and published to GitHub for strict version control. Verification steps confirmed the successful execution of playbooks and operational readiness of both Nginx and Prometheus services across Ubuntu and CentOS environments.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

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Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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